

Extended Abstract

Motivation Citizen-science platforms now supply millions of biodiversity records, yet only a fraction originate from consistently reliable observers whose careful, well-documented submissions shape high-quality species-distribution models. This project seeks to learn the policy of reliable observers using imitations learning, capturing preferences for, e.g., taxonomic novelty, multi-angle photography, and habitat diversity. By focusing exclusively on trajectories from demonstrably reliable users, we aim to (i) construct RL-style trajectories from passive citizen-science logs, (ii) learn a compact policy that captures expert decision pattern.

Method We begin by selecting “power users” from the DivShift-NAWC iNaturalist dataset ($\geq 10,000$ research-grade observations) and partitioning them into train/val/test splits by user. Our trajectory-construction pipeline stitches each user’s timestamped uploads into ordered demonstrations (s_t, a_t) . We derive two alternative state–action encodings: trajectory 1 fuses counts of species and photos in the past seven days, ecoregion diversity, and seasonality with discretized action bins for taxon-rarity and photo count and trajectory 2 replaces ecoregion with daily distance traveled and introduces a “new-taxon” flag alongside photo-count bins. We then apply behavioral cloning with a feed-forward network and transformer policy over a sliding window of past states to predict actions, generate an expert policy, and generate new expert trajectories. For privacy, we aggregate GPS into ecoregion and distance traveled bins and summarize timestamps via rolling-window counters.

Implementation We compare three approaches. First, a naive baseline that simply samples each action according to its empirical frequency in the training set. Second, we implement an MLP for the feed-forward network. Third, a transformer-based policy incorporates a learned [CLS] token (with positional embeddings up to length 50). We assess predictive performance using top-1 and top-3 accuracy on held-out trajectories to gauge how often the agent’s guesses match the expert’s choice. Furthermore, since there are two distinct elements of the action space-rarity and photo count—we also compute top-1 accuracy for each element. Lastly, to measure overall policy fidelity, we compute the Jensen–Shannon divergence between the expert action distribution and the agent’s generated action distribution based on the learned policy.

Results Quantitatively (Tables 3-5), the MLP achieves up to 38.7 % top-1 accuracy on Trajectory 2, 3.9 pp above both the baseline and transformer, and halves the transformer’s Jensen–Shannon divergence (0.198 vs. 0.445). Qualitatively (Figures 3-5), we observe that (i) state and action distributions are heavily skewed toward a single class, (ii) both models converge rapidly but the transformer plateaus at higher loss despite hyperparameter sweeps, and (iii) on held-out data the transformer defaults entirely to the majority action while the MLP shows limited variation.

Discussion Our findings highlight two core challenges. First, class skew drives both agents to trivial policies, with the MLP’s simpler inductive bias proving more robust than the transformer’s capacity. Second, and most importantly, the pseudo-Markovian assumption fails: expert choices depend on long-term motivations and latent context beyond any finite window. By forcing fixed-length state encodings, we discard critical information, handicapping both agents’ ability to recover true decision dynamics.

Conclusion We have demonstrated a streamlined approach for behavioral cloning of citizen-science experts, showing that a lightweight MLP can better capture expert distributions than a heavier transformer under extreme class imbalance. Crucially, we reveal that reconstructing rich, history-dependent policies from passive logs demands richer state representations or explicit reward modeling. Our trajectory pipeline and benchmarks pave the way for future work integrating inverse RL, memory-augmented models, and continuous embeddings: key steps toward scaling expert-level sampling strategies and improving the fidelity of global biodiversity monitoring.

Imitation Learning for iNaturalist Observer Behavior

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Abstract

We present a novel framework for learning expert observer policies from passive, internet-scale biodiversity logs. By converting research-grade iNaturalist submission histories into discrete state–action trajectories, we train both a compact feed-forward network and a transformer policy via behavioral cloning. Our feed-forward model consistently outperforms a naive frequency baseline and the transformer on top-1/top-3 accuracy and achieves roughly half the Jensen–Shannon divergence from expert distributions. Qualitatively, both agents struggle under extreme class skew and the false Markov assumption inherent in fixed-window encoding, defaulting to majority actions. Lastly, we outline directions, such as inverse RL and memory-augmented architectures, for scaling expert-level sampling strategies in citizen science.

1 Introduction

Citizen-science platforms now supply millions of biodiversity records every year, yet only a small fraction originates from consistently reliable observers whose careful, well-documented submissions form the backbone of high-quality species-distribution models (Figure 1). At the same time, a growing body of work has shown how to extract meaningful biodiversity indicators from unstructured, community-contributed data by reverse-engineering survey structure, modeling observation processes, borrowing strength across taxa, and integrating with standardized surveys (Rapacciuolo et al., 2021). However, all of these approaches treat observer behavior primarily as noise to be filtered or modeled out.

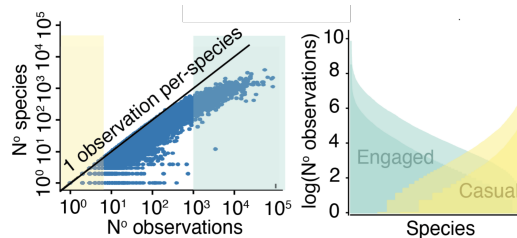


Figure 1: High-quality observers take more observations, allowing for lengthy observation trajectories Sierra et al. (2025)

In this paper, we take a complementary viewpoint: rather than simply correcting for “who saw what where,” we ask whether we can learn how expert observers make their decisions and then replay or simulate them. To that end, we (i) construct trajectories of state–action pairs from passive iNaturalist logs for a cohort of “power users,” (ii) train compact behavioral-cloning agents to predict each user’s next choice, and (iii) assess policy fidelity via standard accuracy metrics and distributional divergence measures.

By modeling the policy of reliable observers, we aim to open new avenues for scaling expert-level data collection strategies across citizen-science platforms. Furthermore, we ask: 1) Can we recover expert observer policies from passive logs? 2) Which model architecture best captures those policies?

2 Related Work

Existing literature notes the unreliability of the iNaturalist "research-grade" label and proposes rule-based confidence scores on iNaturalist observations (Ackland et al., 2024; Campbell et al., 2023). Furthermore, previous work documents how power users and community identifiers lift overall data accuracy (Sierra et al., 2025). Analyses of temporal-spatial participation patterns show that visitor bias and casual engagement skew sampling effort (Di Cecco et al., 2021; Dimson and Gillespie, 2023), yet prior studies stop at descriptive statistics.

On the methodological side, behavior cloning treats imitation as supervised sequence prediction, bypassing the need for an explicit reward signal. Early work such as ALVINN showed that a shallow network trained on steering traces could keep a van running on real roads (Pomerleau, 1989). Recurrent models have been used to forecast human mobility (Lin, 2018), and transformer-based choice models now scale sequence imitation to massive GPS-route datasets (Qiu et al., 2024).

Existing observer-quality studies remain largely descriptive, relying on hand-crafted heuristics and coarse confidence scores, instead of learning the implicit decision policies that drive expert contributions. No prior work casts passive iNaturalist histories as structured demonstrations for learning implicit expert policies. Our work closes this gap by recasting passive observation logs as structured demonstration trajectories and tackling the unique challenge of converting noisy, internet-scale data into state-action pairs amenable to behavior cloning. To do so, we develop a streamlined pipeline that denoises raw uploads, encodes high-dimensional contextual features, and creates trajectories across long horizons. In enabling behavior cloning agents to learn expert observer policies from vast, unstructured citizen-science records, we help explore the need for reinforcement-learning research on internet-scale data (Wang et al., 2023).

3 Method

3.1 Data and Trajectories

From the DivShift-NAWC dataset, I isolate power-users using thresholds on the number of research-grade observations they have (with a minimum of 10,000 observations) and divide them into 70% train, 15% val, and 15% test splits by observers (Sierra et al., 2025).

DivShift-NAWC was collected for ecological analysis, not control or reinforcement learning; it contains no explicit notion of "state" or "action." Our first technical contribution is therefore a *trajectory-construction pipeline* that leverages user IDs to stitch raw, timestamped observations into sequential demonstrations. We (i) derive environment state s_t by fusing spatiotemporal context with session-level running counters, and (ii) discretize each observation into an action a_t that encodes the strategic choice of photographing a taxon of a given rarity with a given photo count. This re-casting of passive citizen-science logs into (state, action) pairs enables imitation learning on a citizen-science platform.

Each reliable observer’s upload history is converted into a chronological sequence of state-action pairs:

$$(s_0, a_0), (s_1, a_1), \dots, (s_T, a_T),$$

where index t denotes the instant an observation is posted. After discussing with iNaturalist data experts, I create two trajectories to see how sensitive model performance is to state-action pair design (Tables 1, 2).

3.2 Behavior Cloning

We use Behavioral Cloning to learn power-users’ policies. We treat action prediction as an multiclass classification problem of the action conditioned on a structured state vector. To do this, we implement a feed-forward neural-network conditioned only on the current state and a transformer model conditioned on a sliding window of previous states to give the model more temporal context.

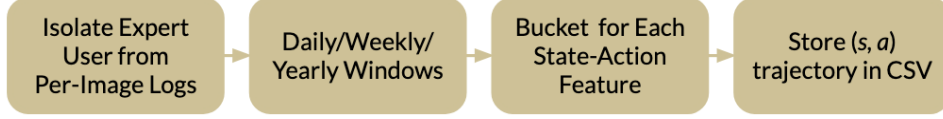


Figure 2: Trajectory Construction Pipeline

Symbol	Encoding (feature vector)
State s_t	• num species in last 7 days (0-1, 2-5, 6-20, 21+) • num photos in last 7 days (0-1, 2-10, 11-50, 51+) • num ecoregions visited in last year (0-1, 2-3, 4+) • summer/winter or both in last year (0-1, 2)
Action a_t	• taxon-rarity (frequent, common, rare) • num photos (0-1, 2-3, 4+)

Table 1: Trajectory 1 State and action representations

Symbol	Encoding (feature vector)
State s_t	• num species in last 7 days (0-1, 2-10, 11-40, 41+) • num photos in last 7 days (0-1, 2-20, 21-100, 101+) • distance traveled today (0-1 km, 1-5 km, 5-20 km, 20+ km) • one-hot summer-winter observation in past year (0-1, 2)
Action a_t	• new-taxon-for-user (0, 1) • num photos (0-1, 2-3, 4+)

Table 2: Trajectory 2 State and action representations

3.3 Trajectory Privacy

iNaturalist observers consent to latitude/longitude and the date of their photos to be shared to assist in species identification. While this data is publicly available, state-action pair construction must not violate the privacy of these users (ex. tracking location patterns over time). To preserve privacy in state-action trajectories, we convert exact GPS coordinates into ecoregion bins and discretized distance-traveled features, omitting all precise latitude/longitude data. Furthermore, exact timestamps are summarized via rolling windows (e.g. “last 7 days” counts) rather than stored directly.

4 Experimental Setup

4.1 Models and Baselines

We compare three approaches. First, a naive baseline that simply samples each action according to its empirical frequency in the training set. Second, we implement feed-forward network (FFN) as an MLP, taking the encoded state vector as input, passing it through two hidden layers of size 128 with tanh activations, and outputting a distribution over the action space. Third, a transformer-based policy embeds each input token into a 128-dimensional space, applies two encoder layers with four attention heads each, and incorporates a learned [CLS] token (with positional embeddings up to length 50). The final [CLS] representation is projected through a linear layer to predict one of the available actions. Both models train with Adam (lr 1e-4, weight decay 1e-5), Cross-Entropy loss, batch size 512, for 5 epochs.

4.2 Evaluation Metrics

We assess predictive performance using top-1 and top-3 accuracy on held-out trajectories to gauge how often the agent’s guesses match the expert’s choice. Furthermore, since there are two distinct elements of the action space-rarity and photo count-we also compute top-1 accuracy for each element. Lastly, we compute the Jensen–Shannon divergence (JSD) between the expert action distribution and the agent’s generated action distribution based on the learned policy, capturing full distributional fidelity beyond mere accuracy.

5 Results

5.1 Quantitative Evaluation

Table 3 shows that on Trajectory 1 the feed-forward MLP obtains a modest but consistent improvement over both the naive baseline and the transformer. Specifically, the MLP’s top-1 accuracy rises to 31.5 % (vs. 29.8 % for both baseline and transformer), and it likewise edges out on top-3 (76.5 % vs. 76.3 %) as well as on the two action-subspace accuracies (rarity: 40.9 % vs. 38.9 %; photo count: 76.4 % vs. 76.3 %).

Table 3: Trajectory 1 Performance (%) Comparison

Method	Top-1 Accuracy	Top-3 Accuracy	Rarity Accuracy	Photo Count Accuracy
Baseline	29.8	76.3	38.9	76.3
MLP	31.5	76.5	40.9	76.4
transformer	29.8	76.3	38.9	76.3

On Trajectory 2 (Table 4), the gap is even larger: the MLP reaches 38.7 % top-1 (baseline/transformer: 34.8 %), a 3.9 pp gain, and boosts rarity accuracy by 5.2 pp (44.1 % vs. 38.9 %), while the transformer only marginally beats the baseline on top-3 and photo-count accuracy.

Table 4: Trajectory 2 Performance (%) Comparison

Method	Top-1 Accuracy	Top-3 Accuracy	Rarity Accuracy	Photo Count Accuracy
Baseline	34.8	83.7	38.9	83.7
MLP	38.7	84.4	44.1	83.8
transformer	34.8	84.1	38.9	83.7

The Jensen–Shannon divergences in Table 5 reinforce this pattern. The MLP yields JSDs of 0.247 and 0.198 on Trajectories 1 and 2 respectively—indicating a close match to the expert action distribution—whereas the transformer’s JSDs (0.495 and 0.445) are roughly double, signifying substantially poorer distributional fidelity.

Table 5: JSD between Behavior-Cloning Generated Trajectories and Expert Trajectories

(a) Trajectory 1		(b) Trajectory 2	
Method	JSD	Method	JSD
MLP	0.247	MLP	0.198
transformer	0.495	transformer	0.445

5.2 Qualitative Analysis

Figure 3 reveals that both the state and action distributions in the training data are extremely skewed: a handful of state bins and a single action class dominate the logs. Such skewness means that both the FFN and the transformer struggle to learn meaningful policies for rare events and instead tend to default to the most common class.

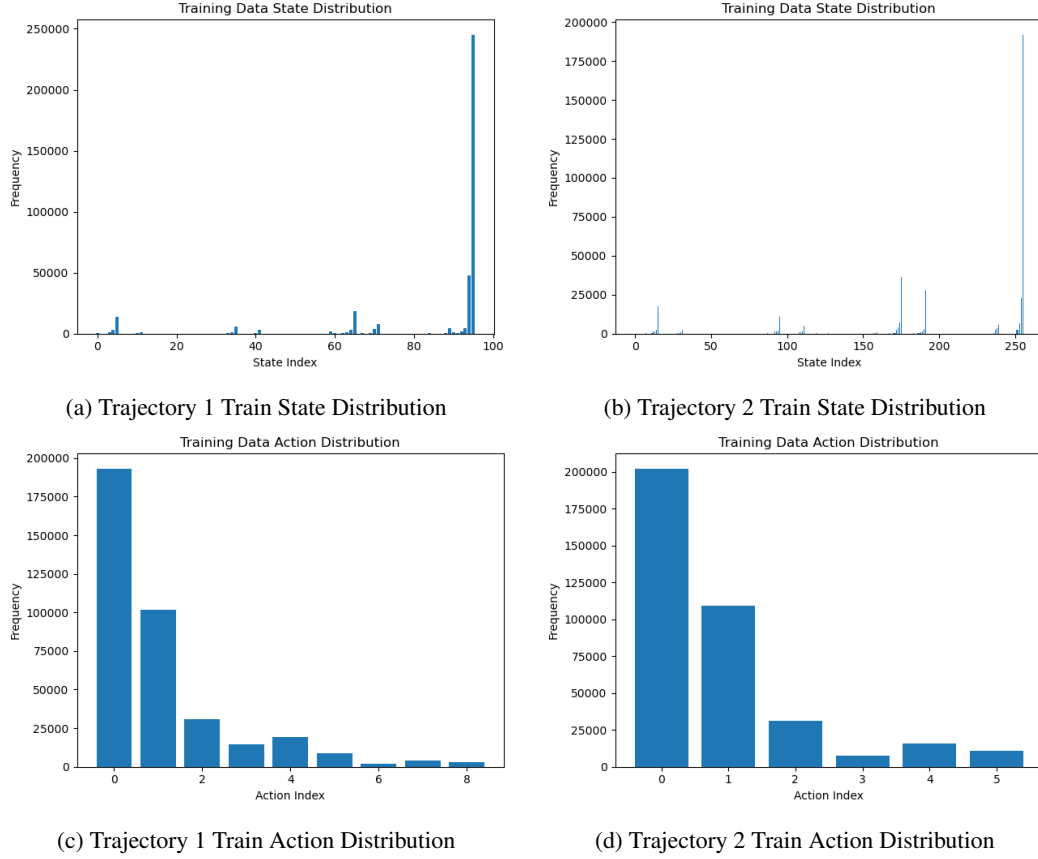


Figure 3: Training Data State and Action Distributions

In Figure 4, we see that both models’ training losses drop rapidly and converge within fewer than three epochs on each trajectory, but the FFN consistently reaches a lower loss plateau than the transformer. Notably, varying the transformer’s window size and learning rate on Trajectory 1 did not alter this outcome: every variant converged to the same suboptimal performance.

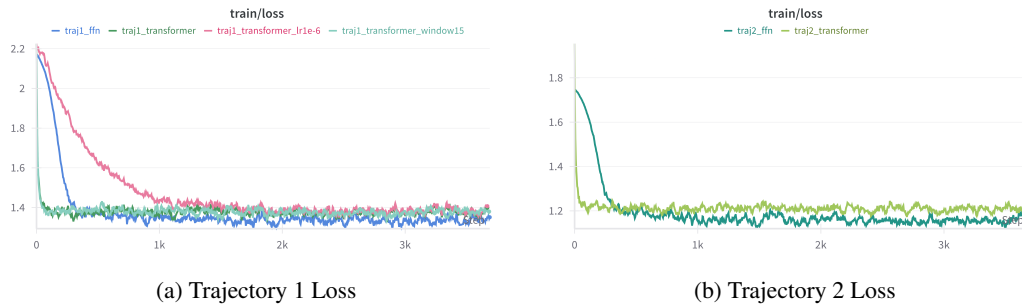


Figure 4: Training Loss Curves

Finally, Figure 5 shows held-out predictions: both agents overfit to the most frequent actions, but whereas the transformer invariably predicts that single majority action, the FFN displays slightly more variation (albeit still heavily biased), underscoring the challenge of learning from such skewed, internet-scale logs

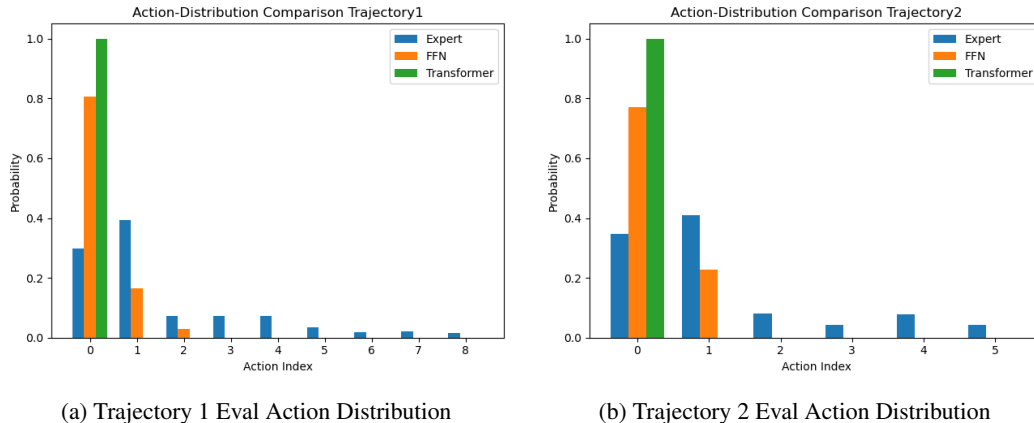


Figure 5: Generated and Expert Trajectory Action Distributions

6 Discussion

Our quantitative and qualitative results reveal several key insights. First, the simple feed-forward MLP consistently outperforms both the naive frequency baseline and the transformer policy on top-1/top-3 accuracy and action-distribution fidelity (Tables 3–5). We attribute this to the extreme skew in state and action distributions (Figure 3), which encourages both models—especially the higher-capacity transformer—to collapse onto the single most common class rather than learn nuanced decision rules. Although the transformer’s attention mechanism can, in principle, leverage longer temporal contexts, we found that varying window size or learning rate did not break its tendency to plateau at suboptimal loss (Figure 4).

Second, both agents struggle on rare events: their predictions on held-out data remain heavily biased toward majority actions (Figure 5). Our hyperparameter sweeps showed no remedy for the transformer’s collapse, highlighting the strength of simple inductive biases under extreme imbalance. This overfitting highlights the limitations of pure behavioral cloning when applied to internet-scale logs without explicit reward signals or imbalance mitigation. Our discretized state–action encoding further simplifies the decision space but may discard subtle cues necessary for modeling expert strategies in low-frequency regimes. Most importantly, however, the underlying assumption that each state–action pair is Markovian fails: in reality, an observer’s next choice depends on unobserved factors (e.g. shifting motivation, external environmental cues, multi-session context) that are not captured by our fixed-length window. By forcing a pseudo-Markovian structure onto rich, history-dependent behavior, both agents are handicapped in recovering the true transition dynamics.

Looking ahead, integrating inverse-reinforcement-learning or off-policy RL could help recover implicit reward functions that better capture the value of rare observations and non-Markovian dependencies. Incorporating latent-variable models or memory-augmented architectures may also address the deep history effects we have thus far ignored. Finally, extending this pipeline to other citizen-science platforms will test its generality and inform methods for scaling expert-level data collection across diverse ecological monitoring tasks. By learning expert policies, this work may help design active-sampling strategies that allocate limited field effort more equitably, potentially reducing taxonomic and geographic bias in global biodiversity monitoring.

7 Conclusion

We have presented a streamlined framework to convert passive iNaturalist logs into structured demonstration trajectories and applied behavioral cloning to learn expert observer policies. Our feed-forward MLP achieved the strongest predictive performance, closely matching expert distributions, while the transformer defaulted to trivial strategies under severe class skew. Crucially, we also identify a fundamental limitation: casting rich observer histories as pseudo-Markovian state–action pairs oversimplifies the decision process and limits both model classes. By releasing our trajectory-construction pipeline and benchmark results, we lay the groundwork for future work that integrates

reinforcement-learning techniques, richer state representations (e.g. latent variables or memory buffers), and approaches explicitly designed for partially observable sequential data. Such advances will be critical for scaling expert-level sampling strategies and ultimately enhancing the quality of global citizen-science biodiversity records.

8 Team Contributions and Changes from Proposal

For the second trajectory design, I consulted CS PhD student Lauren Gillespie to improve from trajectory 1. I contributed all the work presented in this paper

I added a second trajectory to see how differences in trajectory design affect model performance.

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